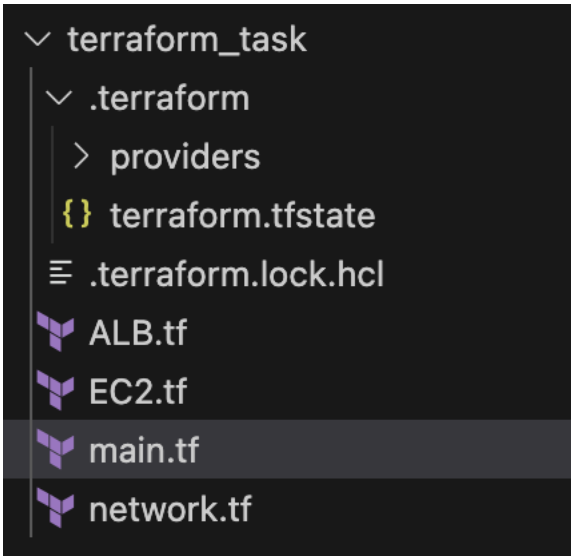
Name – Meenakshi

Batch – **7943**

Project: Launching an nginx web-based application in the three-tier architecture on AWS using Terraform

Tasks:

- Create networking: VPC and Subnets (two public and two private).
- Create an EC2 Machine in the private subnet.
- Create a load balancer in the public subnet, and map it with the EC2 created above using Target groups.
- Use the S3 backend and dynamo DB locking.



ALB.tf

```
resource "aws_lb" "task_lb" {
name = "task-lb-tf"
internal = false
load_balancer_type = "application"
subnet_mapping {
subnet_id = aws_subnet.public1.id
allocation_id = aws_eip.natEip.address
}
```

```
subnet_mapping {
subnet_id = aws_subnet.private1.id
allocation_id = aws_eip.natEip.address
}
}


resource "aws_lb_target_group" "task_TG" {
name = "tf-lb-tg"
port = 80
protocol = "HTTP"
vpc_id = aws_vpc.vpc.id
target_type = "instance"
}

resource "aws_lb_target_group_attachment" "tG_attach" {
target_group_arn = aws_lb_target_group.task_TG.arn
target_id = aws_instance.web.id
port = 80
}

resource "aws_lb_listener" "http" {
load_balancer_arn = aws_lb.task_lb.arn
port = 80
protocol = "HTTP"

default_action {
type = "forward"
target_group_arn = aws_lb_target_group.task_TG.arn
}
}
```

EC2.tf

```
resource "aws_instance" "web" {
ami = "ami-0866a3c8686eaeeba"
instance_type = "t2.micro"
```

```
subnet_id = aws_subnet.private1.id
```

```
tags = {
Name = "task_instance"
}
}
```

Main.tf

```
terraform {
required_providers {
aws = {
source = "hashicorp/aws"
version = "5.77.0"
}
}
backend "s3" {
bucket = "terraformfiles02"
key = "tfStateStore/terraform.tfstate"
region = "us-east-1"
dynamodb_table = "task_table"
}
}

provider "aws" {
# Configuration options
region = "us-east-1"
}
```

Network.tf

```
##VPC
resource "aws_vpc" "vpc" {
cidr_block = "10.100.0.0/16"
}
##subnets
resource "aws_subnet" "public1" {
```

```
vpc_id = aws_vpc.vpc.id
cidr_block = "10.100.1.0/24"

availability_zone = "us-east-1a"

}
resource "aws_subnet" "public2" {
vpc_id = aws_vpc.vpc.id
cidr_block = "10.100.2.0/24"
availability_zone = "us-east-1a"

}
resource "aws_subnet" "private1" {
vpc_id = aws_vpc.vpc.id
cidr_block = "10.100.3.0/24"
availability_zone = "us-east-1b"

}
resource "aws_subnet" "private2" {
vpc_id = aws_vpc.vpc.id
cidr_block = "10.100.4.0/24"
availability_zone = "us-east-1b"

}

##IGW
resource "aws_internet_gateway" "igw" {
vpc_id = aws_vpc.vpc.id

}

##NAT
resource "aws_nat_gateway" "taskNAT" {
allocation_id = aws_eip.natEip.id
subnet_id = aws_subnet.public1.id

# To ensure proper ordering, it is recommended to add an explicit dependency
# on the Internet Gateway for the VPC.
depends_on = [aws_internet_gateway.igw]
}

##EIP
resource "aws_eip" "natEip" {
```

```
domain = "vpc"
}

##route table public

resource "aws_route_table" "rtPublic" {
vpc_id = aws_vpc.vpc.id

route {
cidr_block = "0.0.0.0/0"
gateway_id = aws_internet_gateway.igw.id
}




}

##subnets association public RT
resource "aws_route_table_association" "pub1" {
subnet_id = aws_subnet.public1.id
route_table_id = aws_route_table.rtPublic.id
}
resource "aws_route_table_association" "pub2" {
subnet_id = aws_subnet.public2.id
route_table_id = aws_route_table.rtPublic.id
}



##route table private

resource "aws_route_table" "rtPrivate" {
vpc_id = aws_vpc.vpc.id

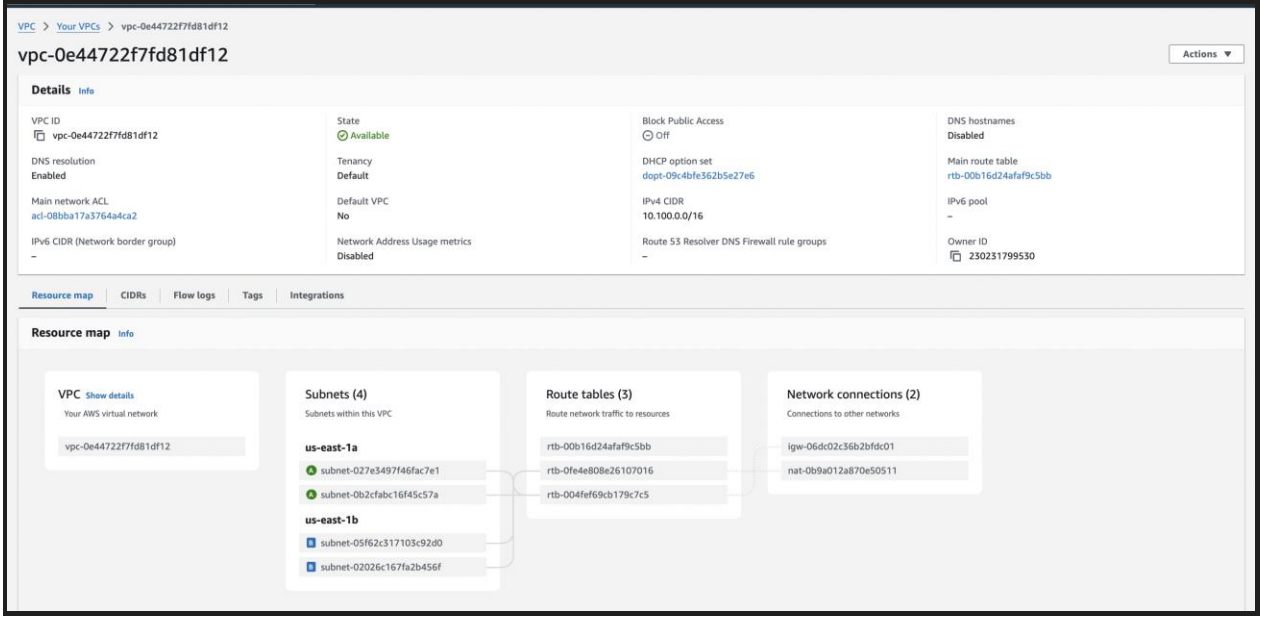
route {
cidr_block = "0.0.0.0/0"
nat_gateway_id = aws_nat_gateway.taskNAT.id
}

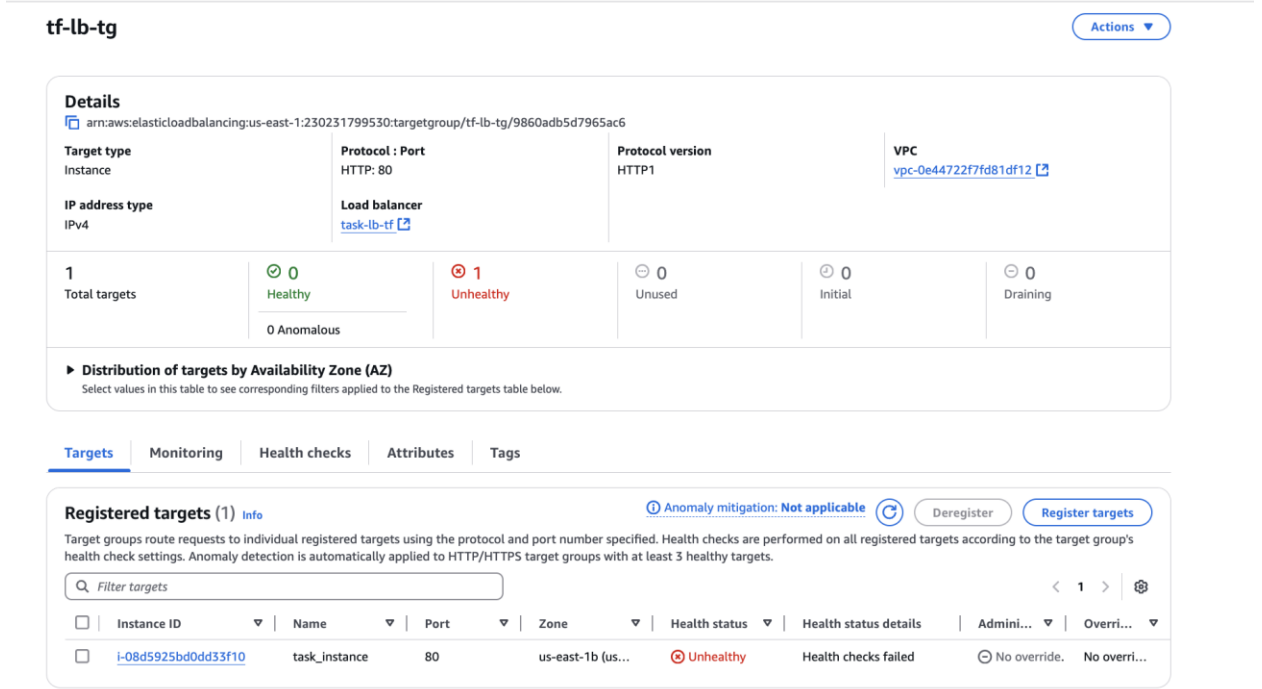


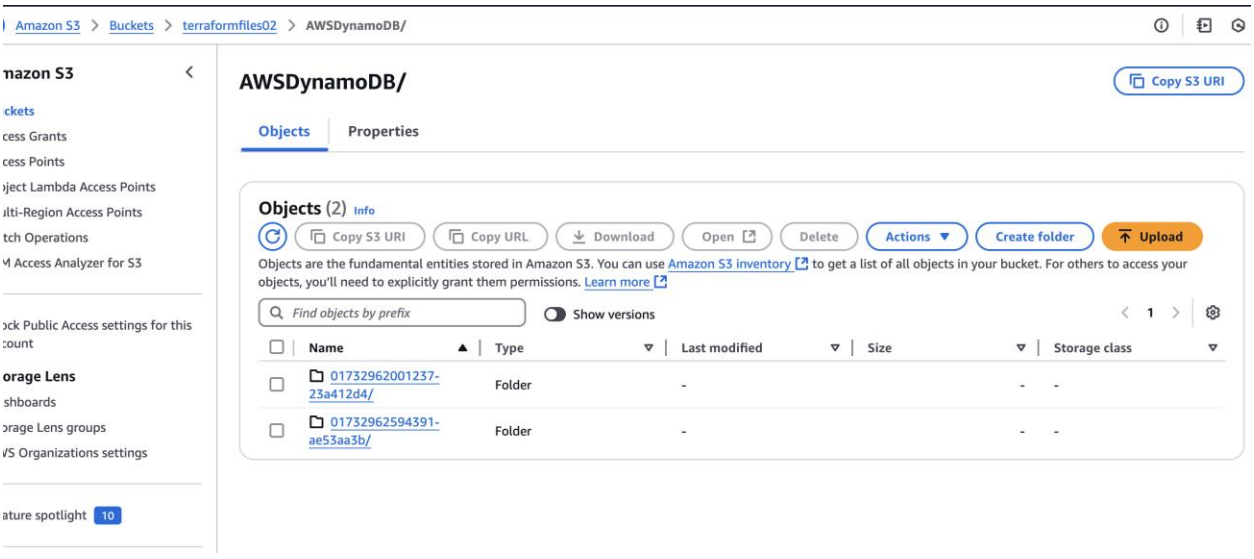
}
##subnets association private RT
resource "aws_route_table_association" "pri1" {
subnet_id = aws_subnet.private1.id
```

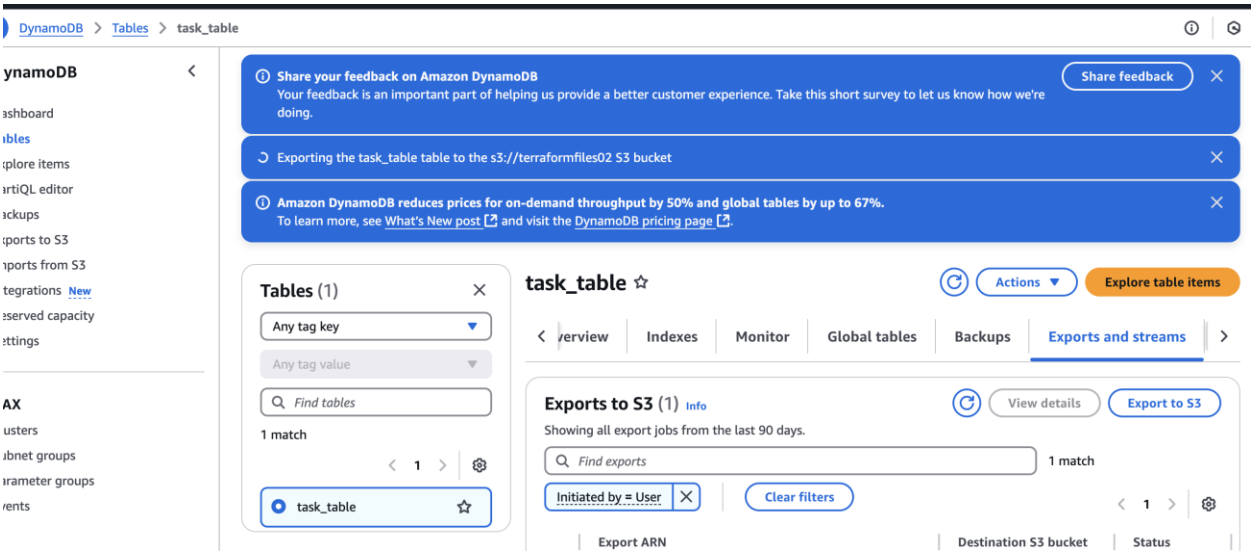
```
route_table_id = aws_route_table.rtPrivate.id
}
resource "aws_route_table_association" "pri2" {
subnet_id = aws_subnet.private2.id
route_table_id = aws_route_table.rtPrivate.id
```

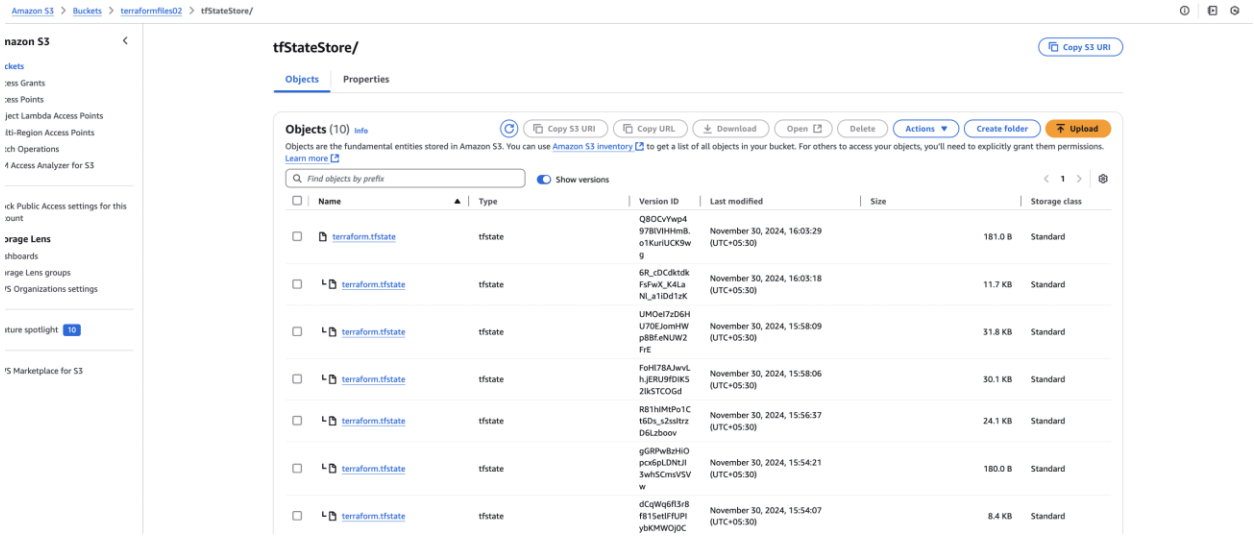
```
Apply complete! Resources: 19 added, 0 changed, 0 destroyed.
```

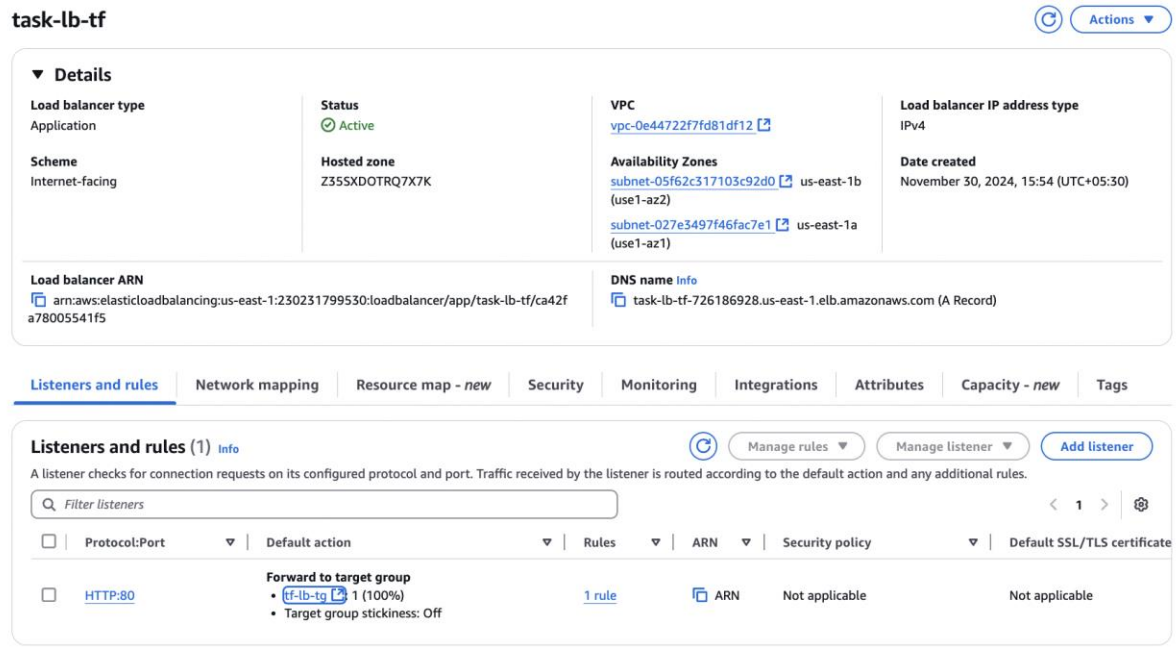





Amazon S3 > Buckets > terraformfiles02 > AWSDynamoDB/
AWSDynamoDB/
Copy S3 URI
Objects
Properties
Objects (2) Info
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Show versions
Name
Type
Last modified
Size
Storage class
01732962001237-23a412d4/
Folder
01732962594391-ae53aa3b/
Folder


DynamoDB > Tables > task_table
Share your feedback on Amazon DynamoDB
Your feedback is an important part of helping us provide a better customer experience. Take this short survey to let us know how we're doing.
Share feedback
Exporting the task_table table to the s3://terraformfiles02 S3 bucket
Amazon DynamoDB reduces prices for on-demand throughput by 50% and global tables by up to 67%.
To learn more, see What's New post and visit the DynamoDB pricing page.
Tables (1)
Any tag key
Any tag value
Find tables
1 match
task_table
task_table
Actions
Explore table items
Indexes
Monitor
Global tables
Backups
Exports and streams
Exports to S3 (1) Info
View details
Export to S3
Showing all export jobs from the last 90 days.
Find exports
1 match
Initiated by = User
Clear filters
Export ARN
Destination S3 bucket
Status


Amazon S3 > Buckets > terraformfiles02 > tfStateStore/
tfStateStore/
Copy S3 URI
Objects
Properties
Objects (10) Info
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions.
Learn more
Find objects by prefix
Show versions
Name
Type
Version ID
Last modified
Size
Storage class
terraform.tfstate
tfstate
November 30, 2024, 16:03:29 (UTC+05:30)
181.0 B
Standard
November 30, 2024, 16:03:18 (UTC+05:30)
11.7 KB
Standard
November 30, 2024, 15:58:09 (UTC+05:30)
31.8 KB
Standard
November 30, 2024, 15:58:06 (UTC+05:30)
30.1 KB
Standard
November 30, 2024, 15:56:37 (UTC+05:30)
24.1 KB
Standard
November 30, 2024, 15:54:21 (UTC+05:30)
180.0 B
Standard
November 30, 2024, 15:54:07 (UTC+05:30)
8.4 KB
Standard


task-lb-tf
Actions
Details
Load balancer type
Application
Status
Active
VPC
vpc-0e44722f7fd81df12
Load balancer IP address type
IPv4
Scheme
Internet-facing
Hosted zone
Z35SXDOTRQ7X7K
Availability Zones
subnet-05f62c317103c92d0 us-east-1b (use1-az2)
subnet-027e3497f46fac7e1 us-east-1a (use1-az1)
Date created
November 30, 2024, 15:54 (UTC+05:30)
Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:230231799530:loadbalancer/app/task-lb-tf/ca42fa78005541f5
DNS name Info
task-lb-tf-726186928.us-east-1.elb.amazonaws.com (A Record)
Listeners and rules
Network mapping
Resource map - new
Security
Monitoring
Integrations
Attributes
Capacity - new
Tags
Listeners and rules (1) Info
Manage rules
Manage listener
Add listener
A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.
Filter listeners
Protocol:Port
Default action
Rules
ARN
Security policy
Default SSL/TLS certificate
HTTP:80
Forward to target group
tf-lb-tg 1 (100%)
Target group stickiness: Off
1 rule
ARN
Not applicable
Not applicable